

Answer Key — Keeping Time with the Skies

Final answers with brief reasons.

Objective

- Q1** (b) **reflecting sunlight** — the Moon reflects sunlight.
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- Q2** (b) **waning** — a shrinking bright part is waning (Krishna Paksha).
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- Q3** (b) **changing Sun–Moon–Earth positions** — phases come from changing relative positions, not Earth's shadow.
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- Q4** (b) **rotation** — Earth's rotation gives the day.
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- Q5** (b) **full Moon** — a lunar eclipse happens at full Moon.
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- Q6** (b) **False** — it rises about 50 minutes later each day.
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- Q7** **29.5** — the phase cycle is about 29.5 days.
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- Q8** **amavasya** — the new Moon day is Amavasya.
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Short answer

- Q9** The Sun lights one half of the Moon; as it revolves around Earth we see different amounts of that lit half, so its bright shape appears to change.
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- Q10** Day: Earth's rotation; month: Moon's phase cycle; year: Earth's revolution around the Sun.
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- Q11** A lunar eclipse is Earth's shadow falling on the full Moon; phases are due to the changing Sun–Moon–Earth positions, not a shadow.
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Long / application

- Q12** At new Moon the Moon is between Earth and Sun, so its sunlit half faces away from us and we see no bright part. At full Moon the Earth is between Sun and Moon, so the whole sunlit half faces Earth and we see a full bright circle. The position relative to the Sun decides how much lit Moon we see.
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- Q13** Earth's rotation gives the day (~24 h); the Moon's phase cycle gives the month (~29.5 days); Earth's revolution gives the year (~365¼ days). Because the year is about a quarter-day more than 365 days, an extra day is added every 4 years (leap year) to keep the calendar in step with the seasons.
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